

Task 2: Fake News Detection using Deep Learning

Objective

Develop a Deep Learning model to classify news articles as Fake or Real.

Dataset

Fake and Real News Dataset

Total Records: 44,898

Technologies Used

- Python
- TensorFlow
- Keras
- Pandas
- NumPy
- Matplotlib

Data Preprocessing

- Dataset merging
- Label encoding
- Missing value checking
- Tokenization
- Sequence padding
- Train-test split

Deep Learning Model

LSTM (Long Short-Term Memory) Neural Network

Architecture:

- Embedding Layer
- LSTM Layer
- Dense Output Layer

Results

Training Accuracy: 99.2%

Validation Accuracy: 98.4%

Test Accuracy: 98.15%

Sample Prediction

Input:

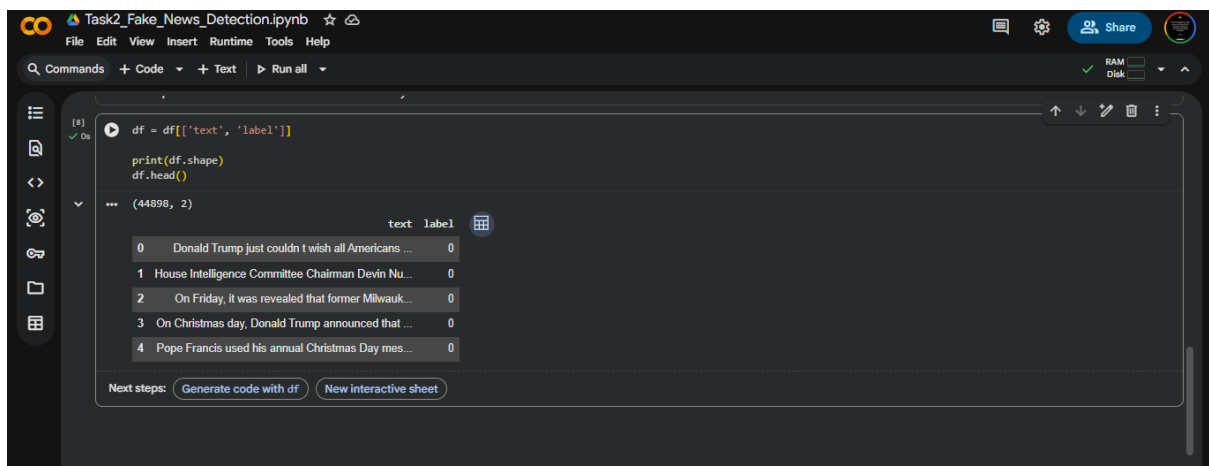
"Scientists discover a new planet that could support life."

Prediction:

Fake News

Conclusion

The LSTM model achieved excellent performance in detecting fake news articles. The project demonstrates the use of Natural Language Processing and Deep Learning techniques for text classification.



The screenshot shows a Jupyter Notebook titled "Task2_Fake_News_Detection.ipynb". The code cell contains the following Python code:

```
df = df[['text', 'label']]
print(df.shape)
df.head()
```

The output shows the DataFrame shape as (44898, 2) and a preview of the first five rows:

	text	label
0	Donald Trump just couldn't wish all Americans ...	0
1	House Intelligence Committee Chairman Devin Nu...	0
2	On Friday, it was revealed that former Milwauk...	0
3	On Christmas day, Donald Trump announced that ...	0
4	Pope Francis used his annual Christmas Day mes...	0

At the bottom, there are two buttons: "Generate code with df" and "New interactive sheet".

Task2_Fake_News_Detection.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
[9]
✓ 0s
print(df.isnull().sum())
```

```
...
text      0
label     0
dtype: int64
```

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
[10]
✓ 0s
df = df.sample(frac=1, random_state=42).reset_index(drop=True)

df.head()
```

		text	label
0		21st Century Wire says Ben Stein, reputable pr...	0
1		WASHINGTON (Reuters) - U.S. President Donald T...	1
2		(Reuters) - Puerto Rico Governor Ricardo Rosse...	1
3		On Monday, Donald Trump once again embarrassed...	0
4		GLASGOW, Scotland (Reuters) - Most U.S. presid...	1

Next steps: [Generate code with df](#) [New interactive sheet](#)

[] Start coding or generate with AI.

Task2_Fake_News_Detection.ipynb

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Commands + Code + Text Run all

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
[20]
✓ 0s
model = Sequential([
    Embedding(input_dim=5000, output_dim=64, input_length=200),
    LSTM(64),
    Dense(1, activation='sigmoid')
])

model.compile(
    loss='binary_crossentropy',
    optimizer='adam',
    metrics=['accuracy']
)

model.summary()
```

Model: "sequential"

Layer (type)	Output Shape	Param #
embedding (Embedding)	?	(unbuilt)
lstm (LSTM)	?	(unbuilt)
dense (Dense)	?	(unbuilt)

Total params: 0 (0.00 B)
Trainable params: 0 (0.00 B)
Non-trainable params: 0 (0.00 B)

Model: "sequential"

Layer (type) Output Shape Param #

embedding (Embedding) ? (unbuilt)

lstm (LSTM) ? (unbuilt)

dense (Dense) ? (unbuilt)

Total params: 0 (0.00 B)
Trainable params: 0 (0.00 B)
Non-trainable params: 0 (0.00 B)

```
[21]
✓ 1m
history = model.fit(
    X_train_pad,
    y_train,
    epochs=3,
    batch_size=32,
    validation_split=0.2
)
```

Epoch 1/3

200/200 — 21s 94ms/step - accuracy: 0.9261 - loss: 0.2095 - val_accuracy: 0.9669 - val_loss: 0.0858

Epoch 2/3

200/200 — 20s 100ms/step - accuracy: 0.9880 - loss: 0.0398 - val_accuracy: 0.9756 - val_loss: 0.0870

Epoch 3/3

200/200 — 19s 95ms/step - accuracy: 0.9922 - loss: 0.0285 - val_accuracy: 0.9844 - val_loss: 0.0513

